

Why Targeted Cooling Beats Citywide Heat Response

Three moves to cool the city where it matters most

Measure hyperlocal risk, cool surfaces, then keep homes survivable through summer.

01

1. Measure Street-Level Heat

Maps reveal zones up to 14 degrees C hotter with 0% to 6% vegetation cover.

02

2. Cool Streets & Surfaces

Cool roofs could save more than \$300 million over the next 35 years.

03

3. Protect Communities

Home cooling for ~500 yearly deaths, 3% of May-September deaths in NYC.



01

Same City, 14 Degrees C Apart: Heat Is Block by Block

Asha Rao maps heat street by street to expose hidden red zones — and where cooling matters most.



Dark lots make this block hotter than the next

Albedo 0-1 explains block gaps: measure reflectivity street by street to target cooling.

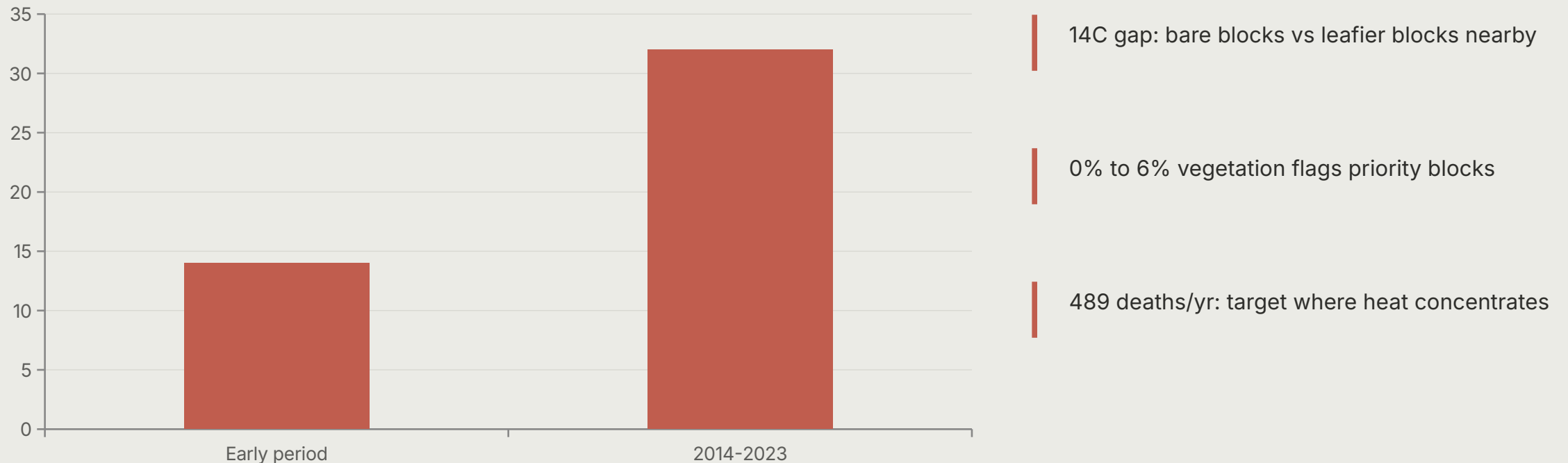
Block heat / albedo

One block runs up to 14C hotter than another as dark roofs, asphalt and 0% to 6% vegetation trap heat; albedo scores reflectivity 0 absorbs to 1 reflects.

Fresh asphalt 0.04 absorbs 96% sun; new concrete 0.55 and grass 0.25 stay far cooler.

Heat doubled unevenly: hottest blocks carry the risk

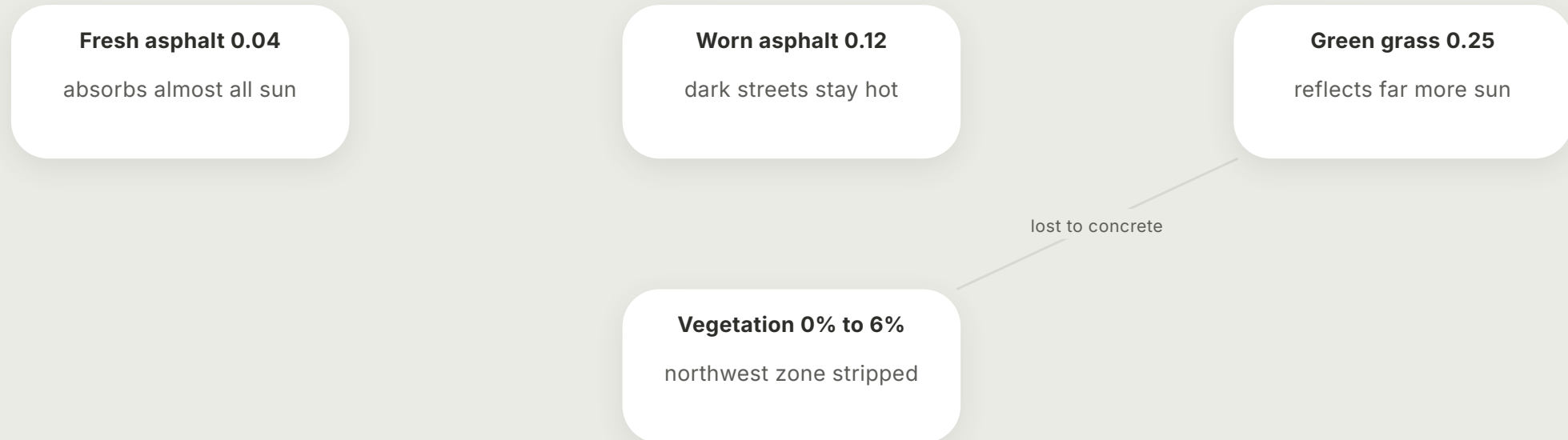
Days over 86F rose 14 to 32 per summer; 14C block gaps show where extra heat lands.





Dark, bare blocks trap heat that feeds on itself

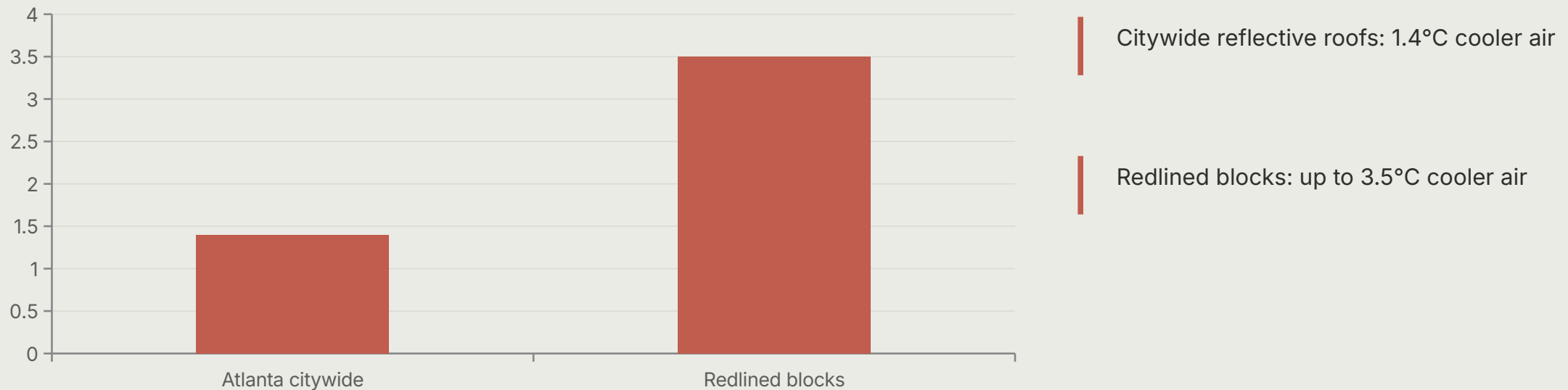
Low-albedo asphalt and 0% to 6% vegetation leave one zone 14 degrees C hotter.



Cool the Surface and You Cool the Street

Cool roofs cut heat most where redlining left it hottest

Atlanta modeling: reflective roofs lower city air 1.4°C, and redlined blocks up to 3.5°C.



Read the two bars left to right: 1.4 then 3.5 — the biggest win lands in the Westside.



Same Street, Two Futures: Trap or Reflect

Fresh asphalt reflects 4% of sun; reflective surfaces, shade and water flip the block from oven to oasis.

BEFORE

Bare asphalt traps the sun

Fresh asphalt albedo 0.04 absorbs 96% of sun; bare metal and no shade bake the block.



AFTER

Cool Reflective Block

New concrete reflects 0.55, with cool roofs, trees and water cooling the block.



No single surface cools the street alone

Reflective roofs cut hotspot air by as much as 3.5 degrees C — shade, pavement and soil lock in that gain.

Tree canopy

Intercepts sun first

Deciduous

0.15 to 0.18

Cool roofs

Reflects heat upward

3.5 degrees C

Reflective

Cool pavement

Cooler ground plane

High albedo

Permeable

Water-wise base

Evaporates stored heat

Moist soil

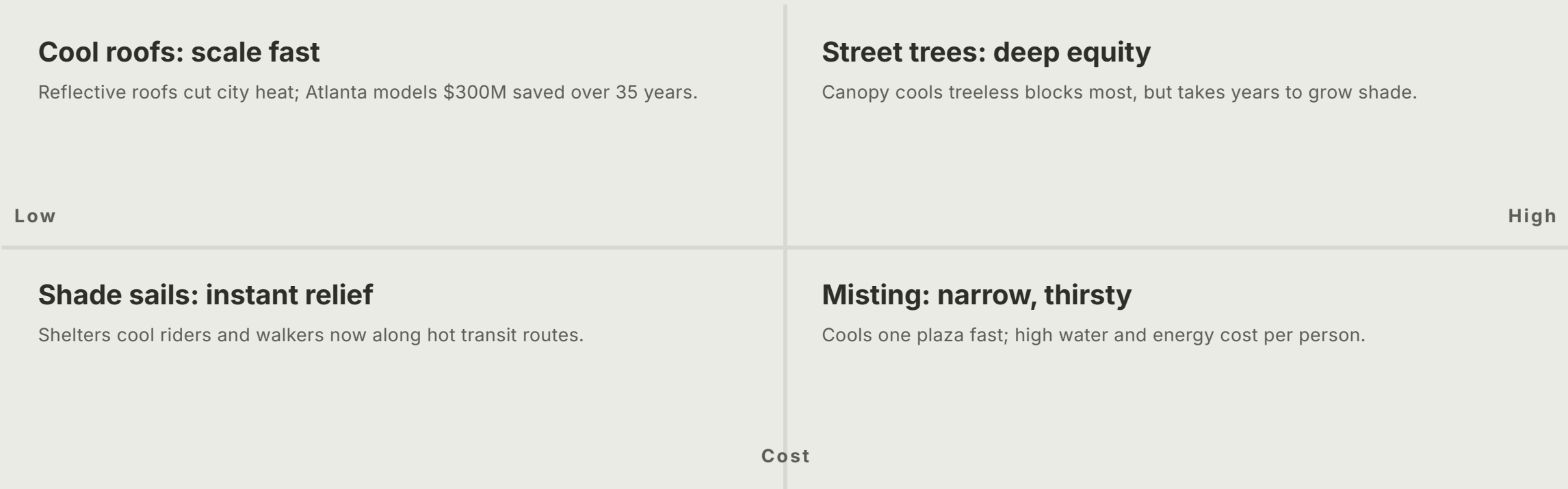
Stormwater



Cool roofs win cheap and fast; trees win big and slow

Place each fix where it pays: fast reflectors for hot roofs, greening where shade inequity runs deepest.

LOW BENEFIT + COOLING + EQUITY HIGH BENEFIT



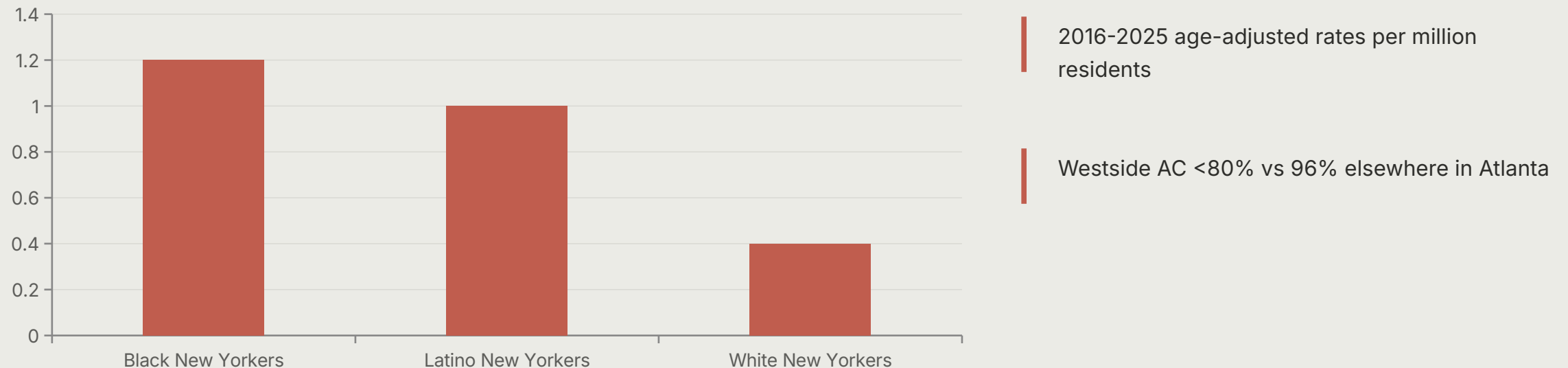
Protect the People Heat Hits First and Worst

Warning Means Nothing Without a Cool Place to Go



Black New Yorkers face 3x the heat death rate

Age-adjusted 1.2 vs 0.4 per million; Westside AC below 80% vs 96% leaves risk at home



If exposure is at home, protection must start at home — not at a distant center.



0 to 36 months: paint now, protect homes, change code

Start with fast reflective roofs, then secure home cooling, then lock in 30% canopy and cool-home law.

	Month 0	Month 6	Month 12	Month 36
0-6 MO: QUICK WINS	Paint roofs reflective Indoor cut 1.5 to 5 degrees C	Open cooling centers Prioritize greatest need areas		Add shade + greenery Relief while trees mature
6-12 MO: COOL ACCESS	Free AC via HEAP Free unit + install if eligible	Ban summer shut-offs No shut-off even if unpaid		Subsidize summer bills Affordable clean energy help
12-36 MO: COOL CODE	Cap indoor summer heat Local Law 23 of 2026	Grow canopy to 30% City goal: 30% tree canopy		Protect strained grid Curb excess office cooling



Check the receipts: where every claim came from

Full source record for mortality, inequity maps, and surface cooling numbers.

NYC Dept. of Health - 2026 NYC Heat-Related Mortality Report, Environment & Health Data Portal.

City heat-death counts and 3x disparity by race used in Section 2.

WRI Ross Center - Cool Cities Lab data platform: hyperlocal heat-risk analysis and scenario modeling.

Land-surface temperature, canopy and cool-roof priority maps.

WRI (Sept. 4, 2026, corr.) - Most Cities Are Getting Hotter. Not All Residents Feel It Equally.

Cape Town, Hermosillo, Atlanta cases; 489,000 deaths/year; cool-roof savings.

Ferreira via WRI - Cape Town to face 20+ more 35C days and 10 more heatwaves/yr by 2050.

Chief Heat Officer projection; Khayelitsha reflective-roof pilot context.



Don't cool the average — cool the block at risk

Measure it, reflect it, get help indoors: that is how 500 preventable deaths end.

About 500 New Yorkers die each warm season, May through September. Measure block heat, reflect it with roofs that cut indoor heat 1.5 to 5 degrees C, grow canopy to 30%, and bring cooling home.

Start with your hottest block this summer